Deformation Theory Seminar $\boxed{12}$ $\boxed{1}$

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# Applications of Bend and Break.

Ref · Debbarc "Higher dimensional algebraic geometry"
· Kollár, Mori "Birational geometry of Algebraic varieties"
· Matsuki "Introduction to the Mori program"
· Cascini "Rational curves on complex manifolds".

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Review of B&B I, II

**B&B I** [D] Prop 3.1 [KM] Cor 1.7

A curve deforming nontrivially while keeping a point fixed (bend)
($\text{i.e } \dim_{[f]} Mor(C, X : f|_{\{c\}}) \geq 1$)

must break into a effective 1-cycle with a rational component passing through the fixed point.

$$\Big( \underset{\text{as limit}}{f_* C \sim f'_* C} + \underset{\text{rational 1-cycle}}{Z} \Big)$$

**B&B II** [D] Prop 3.2 [KM] Prop 1.9.

Starting from a rational curve, it may under certain condition
($\text{i.e } \dim_{[f]} Mor(C, X : f|_{\{0,\infty\}}) \geq 2$)

be broken up into several components (reducible)

Note (Deformation Theory) See [D] Thm 2.6.

- $\dim_{[f]} Mor(C, X) \geq (\text{1-st order}) - (\text{obst.})$
- "def of $f : C \to X$" $\Leftrightarrow$ "emb def of $\Gamma_f \subseteq C \times X$"

see Hartshorne prop 24.9.

then 1-st order $= Ext^0(I/I^2, \mathcal{O}_C)$
obstruction $= Ext^1(I/I^2, \mathcal{O}_C)$

$I$ : ideal sheaf of $i : \Gamma_f \hookrightarrow C \times X$.

We have

$$0 \to I/I^2 \to i^*\Omega_{C\times X} \to \Omega_C \to 0$$
$$i^*\Omega_{C\times X} = \Omega_C \oplus f^*\Omega_X$$

$$\Rightarrow I/I^2 = f^*\Omega_X$$

Hence $\quad Ext^i(I/I^2, \mathcal{O}_C) \cong Ext^i(f^*\Omega_X, \mathcal{O}_C) = H^i(C, f^*T_X)$

So we have $\quad \dim_{[f]} Mor(C, X) \geq \chi(C, f^*T_X)$

$$\overset{R\text{-}R}{=} \deg(f^*T_X) + \operatorname{rank}(f^*T_X)(1-g(C))$$
$$= -K_X . f_*C + (1-g(C)) \dim X.$$

Moreover for $B$: finite, fix $f|_B$.

$$ev_c : Mor(C, X) \longrightarrow X$$
$$f \longmapsto f(c)$$

we want to compute $\dim ev_c^{-1}(f(c))$

$$\Rightarrow \dim Mor(C, X; f|_B) = \dim \ker(H^0(C, f^*T_X) \to T_{f(c)}X)$$
$$\geq -K_X . f_*C + (1-g(C)) \dim X - \#B \dim X$$

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BBI $\quad$ we need $\quad -K_X . f_*C - g(C) \dim X \geq 1.$

BBII $\quad$ we need $\quad -K_X . f_*\mathbb{P}^1 - \dim X \geq 2.$

then it breaks into reduced rational curve $\xi$

until $\boxed{-K_X . \xi \leq \dim X + 1}$

Key : $-K_X . f_*C$ : anti canonical degree (or length)

If $K_X . f_*C \gg 0 \Rightarrow$ deformation alot until it reaches $\dim(X)+1$ rational curve passing through $f(c)$.

First of all we need $-K_X . f_* C > 0$

So we start "Finding rational curves on Fano var"

$\parallel$

$-K_X$: ample

$\Rightarrow -K_X . L > 0$ for $\forall L$: curve.

($\Leftarrow$: unknown see Discord discussion)

<u>Main Thm</u> (Mori) [D] Thm 3.4 or [KM] Thm 1.10.

$X$: Fano, Through any point of $X$, there exists a rational curve of $(-K_X)$ degree at most $\dim X + 1$.

(This says Fano is covered by rational curves (= uniruled)
Moreover, it's known that any two points can be connected by rational curve (= rationally connected $\Longrightarrow$ simply connected)
Campana
See Cascini's survey.)

weaker hypothesis.

<u>Thm</u> (Mori) [D] Thm 3.6 or [KM] Thm 1.13.

$X$: sm proj, $H$: ample, $C$: irred curve s.t $-K_X . C > 0$

$\Rightarrow \exists E \subset X$ rational curve

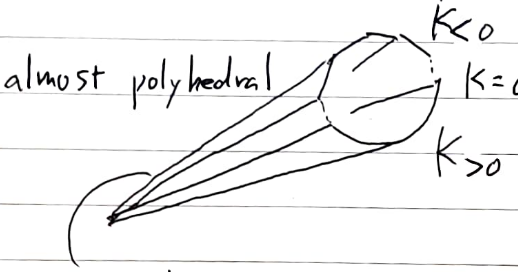
w/ $0 < -K_X . E < \dim X + 1$.

$$\frac{E.H}{-E.K_X} \leq \frac{C.H}{-C.K_X}.$$

(main application = $\begin{cases} \text{Cone thm for sm case} \\ \text{Kawamata length thm.} \end{cases}$)

## Recall (Cone Thm)

$\overline{NE}(X) = \overline{\{\text{eff } 1\text{-cycle}\}}$ Mori cone




Note
$\left( \begin{array}{l} a+b \in R \text{ extremal ray} \\ \quad \Rightarrow a, b \in R \\ \text{So } \sum a_i C_i \in R \Rightarrow C_i \in R \end{array} \right)$

extremal ray $\mathbb{R}[C_i]$
representatives are rational curve
w/ $0 < -K_X . C_i \leq \dim X + 1$

$\boxed{\text{Cone Thm} \longrightarrow \text{contraction Thm} \longrightarrow \text{MMP}}$

For singular case (klt)
we use Non vanishing, Base pt free, rationality etc..
we don't need today's result.

⚠ There is no known proof of this thm
that uses only transcendental method
We only know how to do that in positive char.

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## pf of Thm for Fano

We want to increase the degree of $f$ fixing genus
- $g(C) = 0$ ok
- $g(C) = 1$ we use multiplicity by $n: C \longrightarrow C$
  $\Rightarrow (f \circ n)_* = n^2 \cdot f_*$
- $g(C) \geq 2$ $\nexists$ endomorphism of degree $>1$ by Hurwitz.

◎ Frobenius morphism allows to increase the degree!
$\Rightarrow$ next week.